MATH329 · OPERATIONS RESEARCH

Midterm 472 Solutions

13/04/2026 · 60 minutes · 30 marks

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01 Solution Q1: True / False

#	Answer	Reason
1	**F**	The graphical method works for two variables only, not for more than three.
2	**F**	It started during **World War II**, not the first war.
3	**F**	The objective and all the constraints must be linear, not one or the other.
4	**T**	The tableau organizes the objective and the constraint equations.
5	**F**	The solution must satisfy the constraints, not "may".
6	**T**	This is one of the two requirements to start simplex.
7	**T** *	This is what the answer key shows. See the note below.

A note on item 7

The key shows T. By the lecture, when no variable can enter, the current solution is optimal (**the last solution is optimal**); it does not mean the problem has no feasible solution. So by the lecture the answer is F. The key and the lecture disagree here, so confirm with the instructor.}}

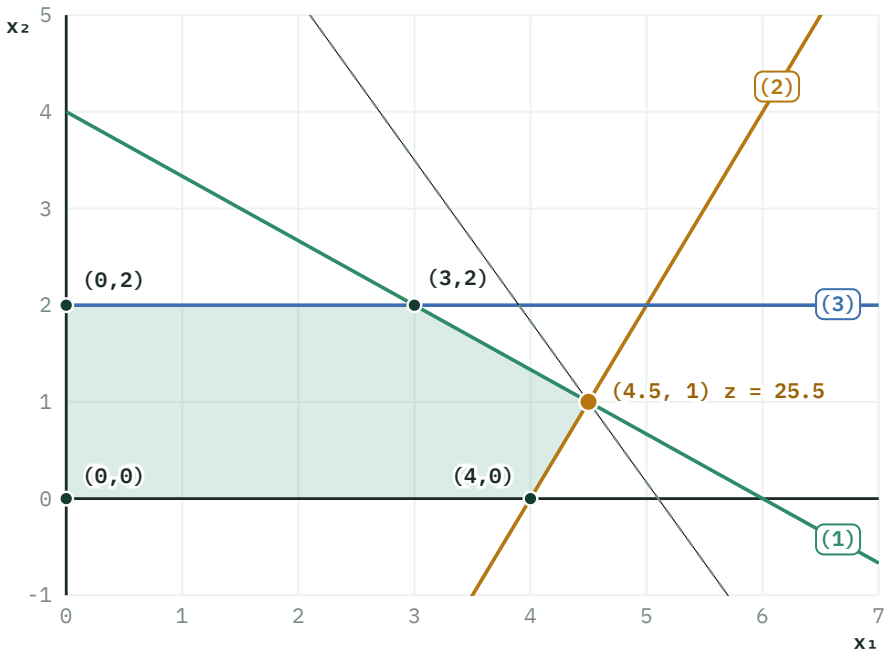
The answer to item 2 is not visible in the photo of the key; F is taken from the lecture.

02 Solution Q2: graphical

1 Prepare the constraints

(1)	$2x_1 + 3x_2 = 12$	points (0, 4) and (6, 0)
(2)	$2x_1 - x_2 = 8$	points (4, 0) and (0, −8)
(3)	$-2x_2 \geq -4$ × (−1) →	$x_2 \leq 2$ horizontal line at $x_2 = 2$

Testing `(0, 0)` satisfies all three, so the region is on the origin's side.



2 The corner points

Corner	Where it comes from	$Z = 5x_1 + 3x_2$
(0, 0)	origin	0
(4, 0)	(2) with the x_1 axis	20
(4.5, 1)	(1) with (2)	**25.5**
(3, 2)	(1) with (3)	21
(0, 2)	(3) with the x_2 axis	6

(1) − (2): $4x_2 = 4 \rightarrow x_2 = 1$ $\quad$ $2x_1 - 1 = 8 \rightarrow x_1 = 4.5$

Objective Value = 25.5 at $x_1 = 4.5$ and $x_2 = 1$

Marks in the key: one for each of the three lines, one for the objective line, two for the solution space, one for the optimum point, and two for the answer.

03 Solution Q3: standard form and simplex

a Standard form 5 M

The second constraint has a negative right-hand side, so multiply it by -1 first:

$$x_1 - x_2 \geq -1 \quad \times (-1) \quad \rightarrow \quad -x_1 + x_2 \leq 1$$

$$\begin{aligned}
&\text{Maximize } Z = 16x_1 + 15x_2 + 0s_1 + 0s_2 + 0s_3 \\
&\text{Subject to} \quad 40x_1 + 31x_2 + s_1 = 124 \\
&\qquad\qquad -x_1 + x_2 + s_2 = 1 \\
&\qquad\qquad x_1 + s_3 = 3 \\
&\qquad\qquad x_1, x_2, s_1, s_2, s_3 \geq 0
\end{aligned}$$

b **Simplex** 10 M · 2 TABLES ONLY

Basic	x_1	x_2	s_1	s_2	s_3	Sol.	Ratio
Z	−16	−15	0	0	0	0	
s_1	40	31	1	0	0	124	$\frac{31}{10}$
s_2	−1	1	0	1	0	1	–
s_3	**1**	0	0	0	1	3	3
Z	0	−15	0	0	16	48	
s_1	0	31	1	0	−40	4	
s_2	0	1	0	1	1	4	
x_1	1	0	0	0	1	3	

The most negative entry in the `Z` row is `−16`, so x_1 enters. The ratios are $\frac{124}{40} = 3.1$ for the first row, none for the second row (negative coefficient), and $\frac{3}{1} = 3$ for the third. The smallest is `3`, so s_3 leaves and the pivot is `1`.

New x_1 row = s_3 row ÷ 1 = (1, 0, 0, 0, 1 | 3)

New Z row = Z row − (−16) × new x_1 row = (0, −15, 0, 0, 16 | 48)

New s_1 row = s_1 row − 40 × new x_1 row = (0, 31, 1, 0, −40 | 4)

New s_2 row = s_2 row − (−1) × new x_1 row = (0, 1, 0, 1, 1 | 4)

We stop here because the question asks for two tables only. The solution is not optimal yet, because `−15` remains in the `Z` row.